

GENERAL SPECIFICATION G5.32

Hermetically Sealed Oil-Filled HV/LV Transformers

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## G5.32 HERMETICALLY SEALED OIL-FILLED HV/LV TRANSFORMERS

### G5.32.1 General

(a) Introduction

- (i) Perform all work; design, supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable Rules, Regulations, Requirements, National and International codes, and Standards as listed in General Specification G5.2- Standards and Regulations.

(b) References

The following are the list of standards which are applicable to this Section:

(i) International Electro-technical Commission (IEC)

| Standard  | Subject                                                       |
|-----------|---------------------------------------------------------------|
| IEC 60060 | High-voltage test techniques.                                 |
| IEC 60076 | Power Transformers.                                           |
| IEC 60214 | Tap-Changers.                                                 |
| IEC 60529 | Degrees of Protection Provided by Enclosures IP Code).        |
| IEC 61558 | Safety of Power Transformers, Power Supply Units and Similar. |

(ii) Singapore Standards

| Standard | Subject                                                                               |
|----------|---------------------------------------------------------------------------------------|
| CP 5     | Code of Practice for Electrical Installation.                                         |
| SS 551   | Code of Practice for Earthing.                                                        |
| SS 516   | Code of Practice for Application, Operation and Maintenance of Dry-Type Transformers. |

- (iii) Conflicts, if any, that may exist between the above reference Standards, Codes and Regulations will be resolved at the discretion of the S.O..

(c) Submittals

- (i) Shop Drawings, and Technical Data: As a minimum, submit the following items:
- 1) Descriptive product information;
  - 2) Itemised bill of material;
  - 3) Dimensional drawings;
  - 4) Operational description;
  - 5) Transformer nameplate data;
  - 6) Anchoring instructions and details;

- 7) Schematic and connection drawings;
- 8) Conduit entrance locations;
- 9) Type Test Certificate;
- 10) Certified Factory Test Reports;
- 11) Manufacturer's installation instructions;
- 12) Operation and Maintenance Manual; and
- 13) Manufacturer's Certification of Proper Installation.

#### **G5.32.2 Products**

(a) Manufacturers

(i) Qualifications

- 1) Minimum of 10 years' experience in the design and manufacture of transformers;
- 2) Certified under ISO 9001 for manufacture of electrical apparatus;
- 3) Manufacturers as listed in the latest edition of the Productivity and Standards Board Directory of SPRING Singapore Certified Products; and
- 4) The product offered shall be Type Tested in accordance with IEC Standards.

(b) General

- (i) Equipment shall be suitable for connection to a 22,000 or 6,600 volts, three-phase, three-wire earthed-wye electrical system;
- (ii) Equipment shall be designed, type tested, and assembled in accordance with standards of IEC 60076;
- (iii) All transformers installed in this Contract shall be from a single manufacturer, and shall be procured directly from the manufacturer;
- (iv) Lifting lugs shall be provided on all equipment and devices weighing over 50 kg;
- (v) Operating Conditions
  - 1) Ambient Temperature: Maximum 40 degrees C;
  - 2) Elevation: < 1000m above Mean Sea Level;
  - 3) Humidity: 100 % RH;
  - 4) Equipment shall be fully rated without derating for these operating conditions; and
  - 5) All transformers shall be energy efficient type and manufactured in compliance with Ecodesign Commission Regulation (EU) No. 548/2014 from the European Commission. Maximum no-load losses, maximum load losses and minimum PEI (Peak Efficiency Index) shall be in accordance with Ecodesign Directive from European Commission.

- (vi) Primary Windings: 22,000 or 6,600 volts, three-phase, three-wire, delta-connected as shown on single-line diagrams. Primary Winding Taps:
  - 1) Fully rated no-load taps (Tap Range +/-5%; Step 2.5%);
  - 2) Two 2-1/2 percent taps above rated voltage;
  - 3) Two 2-1/2 percent taps below rated voltage; and
  - 4) Externally operated no-load tap changer with provisions for locking handle in any tap position.
- (vii) Secondary Windings: 415/240 volts or 433/250 volts at principal tap (as shown on SLD), three-phase, four-wire, wye-connected, solidly earthed with full capacity insulated neutral bushing. Contractor shall verify secondary voltage requirements based on finalized load list and voltage drop calculations;
- (viii) Impedance: As shown on SLD, or in accordance with IEC 60076;
- (ix) Vector Group: Dyn 11;
- (x) Full-height, high voltage terminal;
- (xi) Full-height, low voltage terminal compartment, suitable for cable or busway connection as shown on drawings;
- (xii) Fully enclosed high voltage terminal compartment;
- (xiii) Tank Earthing Pads: One for primary side and one for secondary side;
- (xiv) Base: Suitable for jacking, rolling, or sliding transformer in any direction;
- (xv) Instrumentation
  - 1) Alarm Contacts: 230 volts, inductive, close to alarm; and
  - 2) Three-phase winding temperature detector and dial-type indicator with adjustable alarm contacts.
- (xvi) All Terminals boxes/Junction boxes shall be rated for IP 55 protection.
- (c) Liquid-Filled Transformer
  - (i) Designed, type tested, and assembled in accordance with applicable requirements of IEC 60076.
  - (ii) Hermetically sealed, Mineral Oil immersed type, with barrier between high and low voltage sections.
  - (iii) K-factor: Contractor shall calculate harmonic content based on duty and future loads and design transformer accordingly with the required K-factor.
  - (iv) Self-Cooled
    - 1) Capacity: See single-line diagrams for ratings;
    - 2) Temperature rise by resistance not exceeding 55 degrees C, based on 30 degrees C average ambient over 24-hour period with 40 degrees C maximum; and
    - 3) Top Oil Temperature rise limit shall be 60 degrees as per IEC 60076.
  - (v) Designed to carry short-time emergency overloads in accordance with IEC 60076-7.

- (vi) Primary Insulation Level: 125 kV (peak) BIL at 24 kV, and 75 kV (peak) BIL at 7.2 kV.
- (vii) Sound Rating: 63 dB (A), maximum, for units 1,000 kVA and less. 68 dB (A), maximum, for larger units.
- (viii) Continuous barrel copper coil windings impregnated with thermosetting varnish or epoxy insulation.
- (ix) Core formed from non-ageing laminated silicon steel.
- (x) Tubular radiator coolers, welded to tank wall.
- (xi) Transformer Protection: Transformer shall be protected and necessary control and monitoring shall be provided as specified below:
  - 1) Oil level indicating device with alarm and trip settings;
  - 2) Oil temperature measuring device with alarm and trip settings;
  - 3) Winding temperature measuring device with alarm and trip settings;
  - 4) Pressure release devices with separate alarm and trip contacts;
  - 5) All these protections and alarm & trip contacts shall provide local indication and also go through marshalling panel to SCADA systems; and
  - 6) Necessary RIO shall be provided for communication on TCP/IP MODBUS or PROFIBUS DP protocol for communication with SCADA / PMS as specified in system architecture. Contractor shall provide FNI / RIO's and shall be included in SCADA / PMS regardless of whether they are explicitly mentioned in drawings or I/O list.
- (xii) Welded plate steel tank designated to withstand minimum of 7-1/2 psi pressure with drain valve and sampling device.
- (d) Equipment Identification
  - (i) Nameplate: Diagrammatic, BS EN 10088, Type 306 stainless steel, displaying connection instructions and transformer data.
  - (ii) Warning Sign shall be provided as specified below:
    - 1) Red laminated plastic engraved to white core;
    - 2) To read: DO NOT OPERATE WHEN TRANSFORMER ENERGISED;
    - 3) Mount adjacent to tap changer handle; and
    - 4) Attach with stainless steel screws.
- (e) Factory Acceptance Test
  - (i) Perform production tests on transformers in accordance with IEC 60076-5.
  - (ii) Perform the following Routine Tests in accordance with IEC 60076:
    - 1) Measurement of impedance voltage and load losses;
    - 2) Measurement of no-load losses and no-load current;
    - 3) Measurement of winding resistance and voltage ratio;
    - 4) Testing of vector group conformity;

- 5) Induced overvoltage test;
- 6) Applied voltage withstand test; and
- 7) Dielectric Test.
- (iii) Perform the following Type Tests on single transformer of each rating in accordance with IEC-60076:
  - 1) Partial discharge measurement; and
  - 2) Noise level measurement.
- (iv) Provide Certification of following Type Test / Special Test carried out in accordance with IEC-60076 on a similar rating transformer:
  - 1) Impulse test; and
  - 2) Short Circuit Test.

### **G5.32.3 Execution**

- (a) Installation
  - (i) Install equipment in accordance with manufacturer's instructions and recommendations;
  - (ii) Secure equipment to mounting pads with anchor bolts;
  - (iii) Install equipment plumb and in longitudinal alignment with pad or wall; and
  - (iv) Co-ordinate terminal connections with installation of secondary feeders/busway.
- (b) Site Acceptance Test
  - (i) Site Acceptance Test shall be conducted in accordance with General Specification G5.51 – Electrical Testing.
- (c) Manufacturer's Services
  - (i) Inspection, Commissioning and Site Adjustment: In addition to the requirements of General Specification G2.25 – Manufacturers' Services. An authorised service representative of the manufacturer shall visit Site as required for each transformer to perform the following services:
    - 1) Verify proper installation of the equipment;
    - 2) Inspection, checking and adjusting the equipment; and
    - 3) Commissioning and field testing of the transformer.
  - (ii) Training of operating Personnel: The authorised service representative or a training instructor shall train the operating personnel in all aspects of drive operation and maintenance, including step-by-step troubleshooting procedures with necessary test equipment and in accordance with General Specification G2.27- Training, Advisory and Technical Support.
  - (iii) Provide Manufacturer's Certificate of Proper Installation.

### **END OF SECTION**